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Read the following passage.

Herpes simplex virus (HSV) infects nerve cells in the face, including some near the lips. Like many other viruses, HSV can remain inactive inside the body for years. When HSV becomes active, it causes cold sores around the mouth.

Human cells infected with a virus may undergo programmed cell death. While HSV is inactive inside the body, only one of its genes is transcribed. This gene is the latency-associated transcript (*LAT*) gene that prevents programmed cell death of an infected nerve cell.

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Scientists have found that transcription of the *LAT* gene produces a microRNA. This microRNA binds to some of the nerve cell's own mRNA molecules. These mRNA molecules are involved in programmed cell death of nerve cells. The scientists concluded that production of this microRNA allows HSV to remain in the body for years.

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Use information from the passage and your own knowledge to answer the following questions.

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HSV infects nerve cells in the face (line 1). Explain why it infects **only** nerve cells.

**[3 marks]**

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**[Extra space]**

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**0 9 . 2** HSV can remain inactive inside the body for years (lines 2–3). Explain why this virus can be described as **inactive**.

**[2 marks]**

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**0 9 . 3** Suggest **one** advantage of programmed cell death (line 4).

**[1 mark]**

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**0 9 . 4** The scientists concluded that production of this microRNA allows HSV to remain in the body for years (lines 10–12).

Explain how this microRNA allows HSV to remain in the body for years.

**[4 marks]**

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**[Extra space]**

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**END OF QUESTIONS**

**Turn over ►**